Automation Controller(Formerly known as Ansible Tower)

1. Graphical tool to work with Ansible
2. Easy based on Browser
3. Schedule your play's to run as specific time
4. Logs of plays run
5. RBAC

Control Node/Workstation (Linux/Unix Only) Managing
dnf install ansible-core ansible-navigator python3 -y

Inventory
- Static [static file with system name/IP] (yaml/ini)
- Dynamic [Active Directory, LDAP, Satellite CMDB, Openstack,AWS,VMWare, VirtualBox] via python script

Push a Task to Managed Hosts
1. Ad-Hoc Command (Development/Testing) One time NOT reusable
   ansible hostname -m Module_name -a 'option1=value1 option2=value2'
2. Playbook (Production) Reusable as many time
   ansible-playbook play1.yml

Windows OS
Network Device (cisco v1 v2 v3 v4 modules)
ansible 2.3 2.9

control 1 cisco 1 modules python 2.7
control 2 cisco 2 modules python 3.6

control 2 cisco 2 modules python 3.8

INVENTORY:
Default File: /etc/ansible/hosts [root permission]
Custom File: YOUR_CUSTOM_PATH

ANSIBLE Configuration File:
Default File: /etc/ansible/ansible.cfg Level 1
User file(Home Dir): ~/.ansible.cfg Level 2
Project specific: ./ansible.cfg Level 3
Define Env Variable: ANSIBLE_CONFIG #=/tmp/ansible.config Level 4

Playbooks: Python 3.5
```
vim play.yaml
- name: CISCO IOS
  hosts: list_host
  tasks:
  - name: Install httpd
    ansible.builtin.yum:
        name: httpd
        state: latest
  - name: Create user
    Module_name:
        Module_options:
```

Playbooks: Python 3.6
```
vim play.yaml
- name: Linux Play
  hosts: list_host
  tasks:
  - name: Install httpd
    yum:
        name: httpd
        state: latest
  - name: Create user
    Module_name:
        Module_options:
```

Playbooks: Python 3.9
```
vim play.yaml
- name: Azure Play
  hosts: list_host
  tasks:
  - name: Install httpd
    yum:
        name: httpd
        state: latest
  - name: Create user
    Module_name:
        Module_options:
```

1. Copy module.py from Control Node to Managed Host (Temp Folder)
2. Change permission of module.py to executable via chmod u+x module.py
3. Check & Execute using "python3 module.py" and make changes(if not DESIRED)
4. Remove Module which was copied (rm -rf Module_path)

Container Image1
collection cisco v1
collection mysql v1
python2.7

Container Image2
cisco v2
postgresql
python 3.6

Default Settings for Unix/Linux
ssh.py is plugin
ansible_connection: ssh
ansible_port: 22
ansible_user: student (variable file)
remote_user=devops (ansible.cfg)
ansible_password: redhat (SSH_KEY) [--ask-pass for Ad-hoc]
ask-pass

Custom Settings
winrm.py is plugin
ansible_connection: winrm
ansible_port: 5986
ansible_user: devops (remote_user)
ansible_password: redhat (SSH_KEY)

Custom Settings
ansible_connection: network_cli(cisco,arista)/netconf(juniper)
ansible_port: 22 (Depend)
ansible_user: student (remote_user)
ansible_password: student (SSH_KEY to login)

Idempotent Modules/Behaviour
Desired State (OK, Changed, Fatal, ignoring)
Check if the Managed Hosts has already those configs ?
If Yes, Nothing Happen
If No, Config Changed Done

Managed Hosts/Nodes

Linux
Python Installed
```
[privilege_escalation]
become: true
become_user: root
become_method: sudo/su
become_ask_pass: true/false
```

Windows
PowerShell Installed
```
become: true
become_user: administrator
become_method: runas
become_ask_pass: true/false
```

Network Device
Cisco XE/Juniper/Arista
Python May/May Not
```
become: true
become_user: admin
become_method: ansible.netcommon.enable
become_ask_pass: true/false
```

Cloud

user

In Network Device Case which doesn't have python:
1. Copy Module.py from Control Node to Itself(CN) (Temp Folder)
2. Check & Execute and create commands if changes required
3. Push Commands to Device and Remove Module which was copied (rm -rf Module_path)

Image 1
python 3.5

Container Image1
collection cisco v1
collection mysql v1
python3.5

Image 2
python 3.6

Container Image2
cisco v2
postgresql
python 3.6

Image 3
python 3.9

Container Image3
cisco v2
windows
python 3.9

thread 100 MB
Execution
thread 100 MB
thread 100 MB
thread 100 MB
thread 100 MB
thread 100 MB

OLD Way:
ansible-playbook play.yml -i inventory --ask-become-pass
New Way:
ansible-navigator run linux.yml --inventory gabriel_inventory --eei hub.lab.example.com/ee-supported-rhel8 --pull-policy missing
e 'ansible_become_password=redhat ansible_password=redhat' -m stdout